FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

H-8/4, ISLAMABAD, PAKISTAN — WWW.FBISE.EDU.PK

MATHEMATICS — ANNUAL EXAMINATION 2024

Class: HSSC-II (Class 12) | Time Allowed: 3 Hours | Maximum Marks: 85

Roll No: [ ]

GENERAL INSTRUCTIONS FOR STUDENTS:

1. Answer Section-A on the OMR sheet. 2. Write neat answers for Section-B & C on the answer book.

SECTION — A (Marks: 12 / 17)

Q1. Choose the correct option for each of the following statements:

(i) Which of the following is a fundamental base quantity in SI system?

A) Velocity B) Force C) Electric Current D) Acceleration

(ii) The rate of change of momentum is equal to:

A) Torque B) Applied Force C) Kinetic Energy D) Impulse

(iii) According to Boyle's Law, at constant temperature, pressure is inversely proportional to:

A) Mass B) Volume C) Density D) Molecular weight

SECTION — B (Marks: 33 / 42) — Short Questions

Note: Attempt any ELEVEN parts from the following. All parts carry equal marks (3 marks each).

Q2. (i) Differentiate between vector and scalar quantities with two FBISE syllabus examples.

(ii) Derive the second equation of motion s = vit + 1/2 at^2 using a speed-time graph.

(iii) State Newton's Third Law of Motion. Why do action and reaction forces not cancel each other?

(iv) What is meant by centripetal force? Write down its mathematical formula and SI unit.

(v) Calculate the work done in lifting a 50 kg mass vertically through a height of 10 meters.

(vi) State Pascal's principle and illustrate its application in hydraulic brake systems.

SECTION — C (Marks: 20 / 26) — Descriptive Long Questions

Note: Attempt any TWO questions. All questions carry equal marks (10 marks each).

Q3. (a) State and explain Law of Conservation of Momentum with the help of a mathematical derivation. (6)

(b) A car traveling at 72 km/h decelerates uniformly and comes to rest in 5 seconds. Find distance. (4)

Q4. (a) Describe the working principle and construction of a compound microscope with a neat ray diagram. (6)

(b) Explain total internal reflection and calculate critical angle for glass-air interface. (4)